Cutting SMS Lists

2026-01-09

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
#library(cwi)
if (require(sf)) {
  sf::sf_use_s2(FALSE)
}
```

```
## Loading required package: sf

## Linking to GEOS 3.13.1, GDAL 3.11.4, PROJ 9.7.0; sf_use_s2() is TRUE

## Spherical geometry (s2) switched off
```

```r
library(RSQLite)
library(DBI)
library(leaflet)
library(leaflet.extras2)
library(htmlwidgets)
library(tidygeocoder)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.2.0     v tidyr     1.3.2
## v readr     2.1.6

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(data.table)
```

```
##
## Attaching package: 'data.table'
##
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, mday, minute, month, quarter, second, wday, week,
##     yday, year
##
## The following object is masked from 'package:purrr':
##
##     transpose
##
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
```

```r
library(purrr)
library(leafgl)
library(lubridate)
library(reactable)
library(flextable)
```

```
##
## Attaching package: 'flextable'
##
## The following object is masked from 'package:purrr':
##
##     compose
```

```r
library(magrittr)
```

```
##
## Attaching package: 'magrittr'
##
## The following object is masked from 'package:purrr':
##
##     set_names
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

```r
library(dplyr)
library(randomizr)
```

Read in Data (from other directories)

```r
setwd(r"(C:\Users\Ruairi\Documents\Coding Projects 2025\ct04_testing)")
vf_clean <- read.csv("clean_vf_with_history.csv")

scrubbed_phones_active_dems <- read.csv("scrubbed_numbers 2026-01-03 09-24-35.csv")
scrubbed_phones_active_dems$phone_number <- as.character(scrubbed_phones_active_dems$phone_number)
```

Merge Voterfile with Scrubbed Phones and Limit to Textable Numbers

```r
vf_active_dems_good_phones <- left_join(vf_clean, scrubbed_phones_active_dems, join_by(phone == phone_nu
  filter(fake_number != 'YES') %>%
  filter(phone_line_type == 'CELL PHONE') %>%
  filter(party_code %in% c('D') & status == 'A')

deduped_active_dems_good_phones <-
left_join(vf_active_dems_good_phones,
vf_active_dems_good_phones %>% group_by(phone) %>% summarise(phone_count=n()),
join_by(phone == phone)) %>% filter(phone_count == 1) #%>% arrange(desc(phone_count), phone) #%>% selec

# Only 702 phone numbers attributed to more than 1 person (1,444 people total)
# I drop them above
```

Random Assignment to Message Groups with Balance on Age and Gender

```r
tier_1_towns <- c('New Haven','Hamden','West Haven','Milford','Stratford')

deduped_active_dems_good_phones <-
  deduped_active_dems_good_phones %>%
    mutate(tier1 = ifelse(town_name %in% tier_1_towns, 1, 0)) %>%
    mutate(age_bracket = cut(age,
                             breaks = c(18, 35, 50, 65, Inf),
                             labels = c("18-34", "35-49", "50-64", "65+"),
                             right = FALSE))

lower_tiers <- deduped_active_dems_good_phones %>% filter(tier1 == 0)

set.seed(2026)
lower_tiers$strata <- paste(lower_tiers$gender, lower_tiers$age_bracket, sep = "_")
lower_tiers$message <- block_ra(blocks = lower_tiers$strata,
                                conditions = c("local_choice", "local_voice", "local_dem", "local_soft"
                                               "dist_choice", "dist_voice", "dist_dem", "dist_soft"))
```

Balance checks

```r
#table(lower_tiers$strata, lower_tiers$message)

table(lower_tiers$message, lower_tiers$age_bracket)
```

```
##
##                18-34 35-49 50-64  65+
##   local_choice  1200   936   621  521
##   local_voice   1202   936   620  521
##   local_dem     1200   937   621  521
##   local_soft    1200   937   619  520
##   dist_choice   1203   936   619  522
##   dist_voice    1200   937   620  521
##   dist_dem      1202   936   619  521
##   dist_soft     1200   936   620  521
```

```r
table(lower_tiers$message, lower_tiers$gender)
```

```
##
##                   F    M    U
##   local_choice 2002 1225   51
##   local_voice  2002 1226   51
##   local_dem    2002 1226   51
##   local_soft   2001 1224   51
##   dist_choice  2002 1226   52
##   dist_voice   2002 1225   51
##   dist_dem     2002 1225   51
##   dist_soft    2001 1224   52
```

```r
table(lower_tiers$message, lower_tiers$town_name)
```

```
##
##                Ansonia Beacon Falls Bethany Branford Derby Durham East Haven
##   local_choice     204           40      62      315    77     39        292
##   local_voice      191           56      59      292    73     46        230
##   local_dem        202           35      45      313    71     75        246
##   local_soft       192           58      40      315    89     50        245
##   dist_choice      195           52      45      332    77     43        275
##   dist_voice       213           43      43      313    82     47        279
##   dist_dem         193           48      60      304    68     50        247
##   dist_soft        195           47      49      301    69     64        273
##
##                Guilford Middlefield Middletown Naugatuck North Branford
##   local_choice      221          31        277       282             83
##   local_voice       239          33        302       271             90
##   local_dem         277          38        278       268             88
##   local_soft        280          26        314       255             77
##   dist_choice       241          26        263       287             98
##   dist_voice        255          34        253       255             75
##   dist_dem          281          43        251       263             83
```

```
##   dist_soft         249           23         270       301              77
##
##                North Haven Orange Prospect Seymour Shelton Wallingford
##   local_choice         185    139       53     126     234         341
##   local_voice          182    131       63     142     249         350
##   local_dem            185    117       53     145     276         311
##   local_soft           184    120       57     128     230         348
##   dist_choice          190    118       72     110     232         331
##   dist_voice           199    119       57     137     257         348
##   dist_dem             176    130       66     111     271         327
##   dist_soft            178    141       54     123     261         320
##
##                Waterbury Woodbridge
##   local_choice       175        102
##   local_voice        187         93
##   local_dem          156        100
##   local_soft         163        105
##   dist_choice        194         99
##   dist_voice         182         87
##   dist_dem           186        120
##   dist_soft          185         97
```

```
counts <- lower_tiers %>%
  group_by(strata, message) %>%
  summarise(count = n(), .groups = "drop") %>%
  pivot_wider(names_from = message, values_from = count, values_fill = 0)

col_sums <- counts %>%
  select(-strata) %>%       # exclude the strata column
  summarise(across(everything(), sum)) %>%
  mutate(strata = "Total")  # add a label for the total row

# Bind totals to the table
assignment_table_with_totals <- bind_rows(counts, col_sums)

assignment_table_with_totals %>% head(13)
```

```
## # A tibble: 13 x 9
##    strata  local_choice local_voice local_dem local_soft dist_choice dist_voice
##    <chr>          <int>       <int>     <int>      <int>       <int>      <int>
##  1 F_18-34          762         763       762        762         763        762
##  2 F_35-49          575         575       575        575         575        575
##  3 F_50-64          371         370       371        371         370        371
##  4 F_65+            294         294       294        293         294        294
##  5 M_18-34          421         422       421        421         422        421
##  6 M_35-49          350         350       351        350         350        351
##  7 M_50-64          240         240       240        239         239        239
##  8 M_65+            214         214       214        214         215        214
##  9 U_18-34           17          17        17         17          18         17
## 10 U_35-49           11          11        11         12          11         11
## 11 U_50-64           10          10        10          9          10         10
## 12 U_65+             13          13        13         13          13         13
## 13 Total           3278        3279      3279       3276        3280       3278
## # i 2 more variables: dist_dem <int>, dist_soft <int>
```

```r
lower_tiers %>% group_by(message) %>% summarise(avg_age = mean(age), med_age = median(age))
```

```
## # A tibble: 8 x 3
##   message      avg_age med_age
##   <fct>          <dbl>   <dbl>
## 1 local_choice    43.9      41
## 2 local_voice     44.0      41
## 3 local_dem       43.8      41
## 4 local_soft      43.8      41
## 5 dist_choice     43.9      41
## 6 dist_voice      43.9      41
## 7 dist_dem        43.8      40
## 8 dist_soft       43.9      41
```